Task Title: Description of Different Types of Propositional Operator

Code (C++ Language)**:**

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
int main()
{
    vector<str>S= {"p","q","!p","!q","p/\\q","p\\/q","p->q","p<->q"};
    for(auto m:S)
    {
        cout<<m<<"\t";
    }
    newl;
    for(int p=0; p<2; p++)
    {
        for(int q=0; q<2; q++)
        {
            p?cout<<"T":cout<<"F"; // Just p
            cout<<"\t";

            q?cout<<"T":cout<<"F"; // Just q
            cout<<"\t";

            !p?cout<<"T":cout<<"F"; // Not p
            cout<<"\t";

            !q?cout<<"T":cout<<"F"; // Not q
            cout<<"\t";

            p&&q?cout<<"T":cout<<"F"; // p /\ q
            cout<<"\t";

            p||q?cout<<"T":cout<<"F"; // p \/ q
            cout<<"\t";

            !p||(p&&q)?cout<<"T":cout<<"F"; // p -> q (p implies q)
            cout<<"\t";

            (!p||(p&&q))&&(!q||(q&&p))?cout<<"T":cout<<"F"; // p <-> q (p->q and q->p)
            cout<<"\t";

            newl;
        }
    }
}
```

Output:

```
p       q       !p      !q      p/\q    p\/q    p->q    p<->q
F       F       T       T       F       F       T       T
F       T       T       F       F       T       T       F
T       F       F       T       F       T       F       F
T       T       F       F       T       T       T       T

Press any key to continue . . .
```

Code (Python Language)**:**

```
from prettytable import PrettyTable

table=PrettyTable()

table.field_names =[" p "," q "," !p "," !q "," p /\ q "," p \/ q "," p -> q "," p <-> q "]

for p in range(0,2):
    for q in range(0,2):
        table.add_row([
            bool(p), # Just p
            bool(q), # Just q
            bool(not p), # not p
            bool(not q), # not q
            bool(p and q), # p /\ q
            bool(p or q), # p \/ q
            bool((not p) or (p and q)), # p->q (p imples q)
            bool(((not p) or (p and q)) and ((not q) or (q and p)))]) # (p->q) /\ (q->p)

print(table)
```

Output:

```
+-------+-------+-------+-------+----------+----------+----------+-----------+
|   p   |   q   |   !p  |   !q  |  p /\ q  |  p \/ q  |  p -> q  |  p <-> q  |
+-------+-------+-------+-------+----------+----------+----------+-----------+
| False | False |  True |  True |  False   |  False   |   True   |    True   |
| False |  True |  True | False |  False   |   True   |   True   |   False   |
|  True | False | False |  True |  False   |   True   |  False   |   False   |
|  True |  True | False | False |   True   |   True   |   True   |    True   |
+-------+-------+-------+-------+----------+----------+----------+-----------+
Press any key to continue . . .
```

Discussion:

Proposition expresses the statement either TRUE or FALSE. Here I have used different types of propositional logic and implemented them to know how they work. The logic ‘implies’ and ‘bidirectional’ terms are new for me. First of all I have used c++ language to get the understandings of the program. After that I have written it in Python language to get the table-wise output.